LabLens — understand your bloodwork

Photograph a lab report. It reads the numbers, flags what is out of range, and explains each one in plain English — grounded in reference ranges, never a diagnosis.

AI for Human Health | Vision + grounded reasoning | Working prototype

The problem

People receive lab results as a page of numbers and abbreviations and are told to "discuss it with a doctor" — weeks later. They cannot tell which value matters, what it even is, or whether a flag means anything. Health literacy is one of the strongest predictors of outcomes, and a lab report is exactly where it is tested. The dangerous failure is not confusion; it is **false reassurance** from a "normal-looking" table.

What it does

- **Reads** a photographed report into structured values (vision model, transcription only — unreadable values are named, never guessed).
- **Flags** each value against the reference range **printed on the report itself**, falling back to a curated knowledge base only when the sheet omits one — and always states which range was used.
- **Explains** every out-of-range value in plain language from a hand-built knowledge base: what the test is, what a persistent high or low can mean, and why it matters.
- **Warns honestly:** values crossing a published critical threshold raise a distinct "discuss promptly" banner, never hidden in the table.
- **Trends** multiple reports, joining an analyte across labs that abbreviate it differently.

Why it is different

The model transcribes; a knowledge base interprets; the report's own ranges win. That separation makes a plausible-but-wrong medical claim structurally hard — the opposite of a chatbot asked to "read my blood test". It also says what it could not read and shows its range source, so a user can challenge it.

Verified result (local run)

Metric	Result
Values transcribed from a photographed panel	20 of 21 rows in ~13 s on CPU
Matched to the knowledge base	19
Flagged out of range (ground truth: 11)	11 — 6 high, 5 low; 9 in range
Range source	the report's own printed ranges
Browser console errors	0

Architecture

Photo → **extract** (vision transcription) → **analyze** (the only interpreter: flag vs report range, attach grounded explanation, critical-value banner) → **knowledge base** (26 curated analytes) → Flask UI. Runs on CPU against a local OpenAI-compatible vision endpoint.

Honest limits

It explains, it does not diagnose — it is not a medical device. Reference ranges differ by lab, sex, and age; the report's own range is used when present. The knowledge base covers 26 common analytes; unmatched tests are shown but not interpreted, and named. Trends need more than one report.

Runs locally. `pip install -r requirements.txt && ./run.sh` → http://127.0.0.1:5091. MIT licence.